MIT 18.06 SC - Unit I Exam

1)

a) $r = n < m = 3$ and $r, n \neq 0$

b) The only solution is the 0 vector (with appropriate dimensions) because A is full column rank (i.e. no free variables)

c) Two possible examples are :-

$A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ if $n = 1$ $\quad A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ if $n = 2$

a)

2) $A^{-1} = \cancel{A_{23}}\, E_{23} E_{31} E_{21}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \times \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{bmatrix}$$

b) $A = (E_{23} E_{31} E_{21})^{-1}$

$= E_{21}^{-1} E_{31}^{-1} E_{23}^{-1}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix}$$

c) $L = (E_{31} E_{21})^{-1}$

$= E_{21}^{-1} E_{31}^{-1}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

3)

a) $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 3 & c & 2 & 8 \\ 0 & 0 & 2 & 2 \end{bmatrix}$

$\downarrow$

$U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & c-3 & -4 & -4 \\ 0 & 0 & 2 & 2 \end{bmatrix}$

If $c=3$ then $c-3=0$, and we will have only 2 pivots (in columns 1 and 3) so $C(A) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ \cancel{-4} \\ 2 \end{bmatrix}$

If $c \neq 3$ then $c-3 \neq 0$ and so we will have 3 pivots (in columns 1, 2 and 3) so $C(A) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ c \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

b) If $c=3$ then from part a we have $U = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 2 & 2 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

~~Ø~~ The free variables are hence x_2 and x_4, setting each to 0 while the other to 1 gives us the ~~two~~ following two special solutions:- $[-1 \quad 1 \quad 0 \quad 0]^T$ and $[-2 \quad 0 \quad -1 \quad 1]^T$, and so we have:-

$$N(A) = d_1\begin{bmatrix}-1\\1\\0\\0\end{bmatrix} + d_2\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix}$$

where d_1 and d_2 are any real numbers.

If $c \neq 3$ then we have 3 pivots and x_4 is the only free variable. Solving $Ax = 0$ for $[x_1, x_2, x_3, 1]^T$ we get $x = [-2 \quad 0 \quad -1 \quad 1]^T$, thus we have:-

$$N(A) = d\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix}$$ where d is any real number

~~c) If $c=3$ then the free variables are x_2 and x_4, which we set to 0 and thus solving also if we apply the row operations we did on A to get it into rref to c, we get the vector $b = [1 \quad c \quad 0]^T$ to get $b' = \begin{bmatrix}1\\0\\0\end{bmatrix}$~~

c) $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = x_p$ (since $[1 \ c \ 0]^T$ is its second column).

Now if $c = 3$ then the complete solution is:-

$$x_c = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d_2 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

If $c \neq 3$ then the complete solution is:-

$$x_c = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

4)

a) The nullspace is at least two dimensional.

b) The first, ~~sec~~ fourth and fifth columns form a basis for $C(A)$ which is basically all of R^3. The third column of A is 0, and the second column is $4 \times$ the first column.

c) It is the set of 3×3 upper triangular matrices